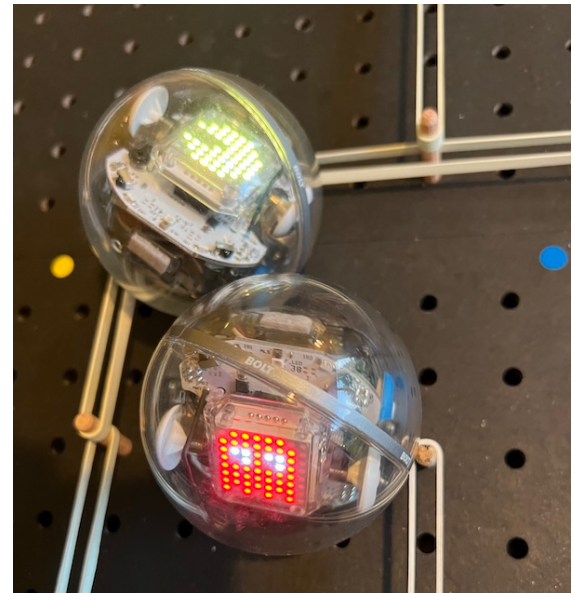




Robot Pac-Man

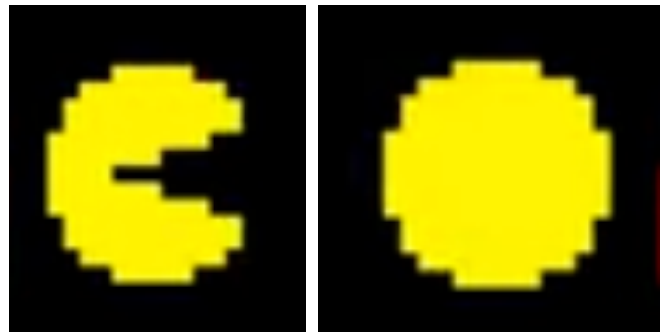
You've (probably) played Pac-Man the video game, but have you played it with robots?



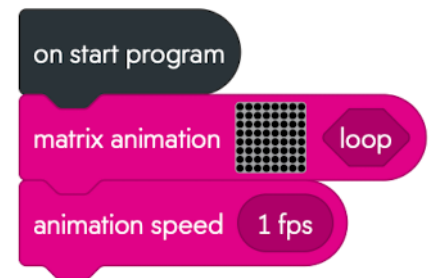
This is the original Namco Pac-Man videogame from 1980. Can we recreate it with robots?

https://youtu.be/i_OjztdQ8iw

*Pac-Man explores a maze, avoiding ghosts. An **energizer** temporarily reverses the chase where ghosts enter a frightened state and become vulnerable.*



- 1) In Chrome, open <https://edu.sphero.com/code>
- 2) Connect your **Sphero BOLT** robot: click 
- 3) Select  then 
- 4) Your BOLT has a sticker with a code on it. Choose the matching BOLT from the list then 
- 5) Check your robot is showing the right image then 
- 6) Click **+ CREATE PROGRAM**. 
- 7) Name your program after the character you're creating. Leave **BLOCKS** selected. Select **SPHERO BOLT**, **deselect SPHERO BOLT+** then **CREATE**.
- 8) Add code for a **matrix animation** with your *Pac-Man* design. Animate it to make its mouth open and close.



Click on the matrix grid to edit then  to add a new 8x8 matrix.

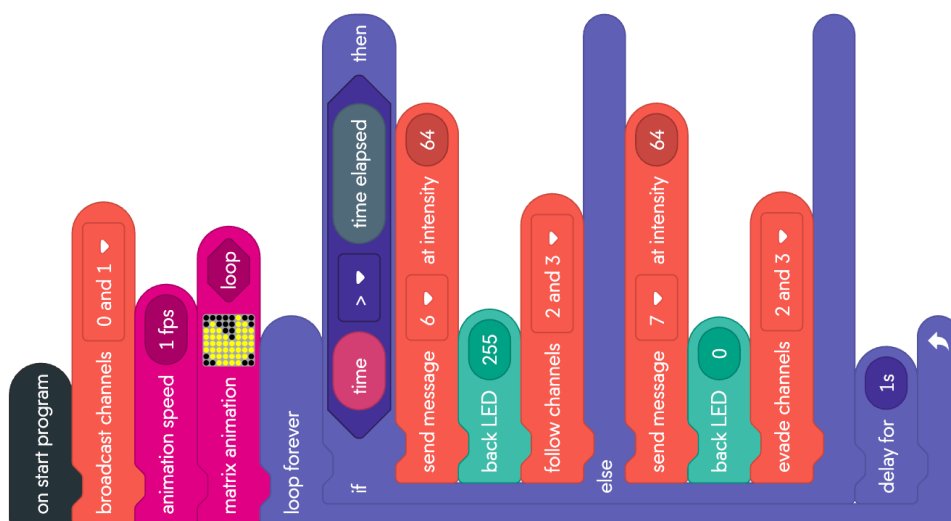
*Click  to finish. Select your graphic then click  to edit it. Click **+** to add a new frame.*

To remove a frame, click , select a frame then .

- 9)  your code. Does it work?
- 10) In **Variables**, create a new variable called **time**.
This is a **number** because it records time elapsed.
We'll use time for the **energizer** later.

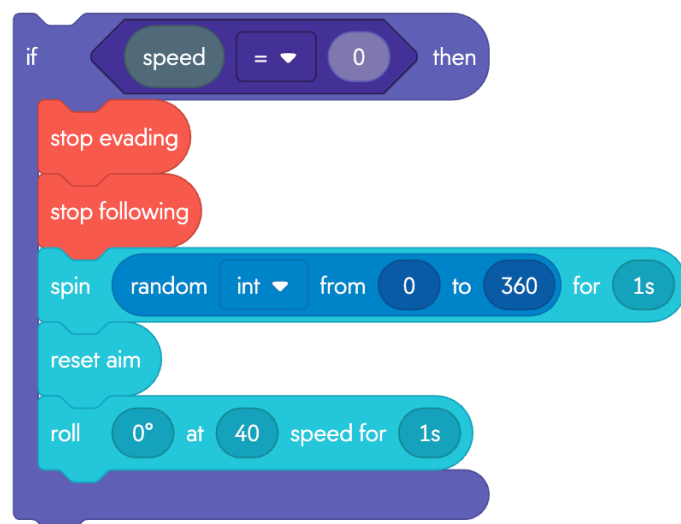
Sphero robots communicate using infra-red light invisible to us. Ghosts broadcast their position on channels 2, 3 (short & long range), and follow Pac-Man on channels 0, 1.

- 11) Replace the existing code with the following:



Sometimes Pac-Man gets stuck on the maze walls.

- 12) Add the code (right) at the end of the loop (but still inside the loop) to detect when it has stopped moving, give it a random spin and roll forwards.



Energizer

In the videogame, the energizer is a large “Power Pill” that Pac-Man can eat to power-up. In the robot game we will use a torch to energize Pac-Man. When we walk from bright sunshine into a house, we can’t see anything until our eyes adjust to the dark. The robot is the same. It has a light sensor, and we measure the average light level. If the brightness suddenly increases, then assume that’s a flash of light from the torch. The light sensor is next to the front LED, so set this to a really dim red (red=1) to show people where to point the torch.

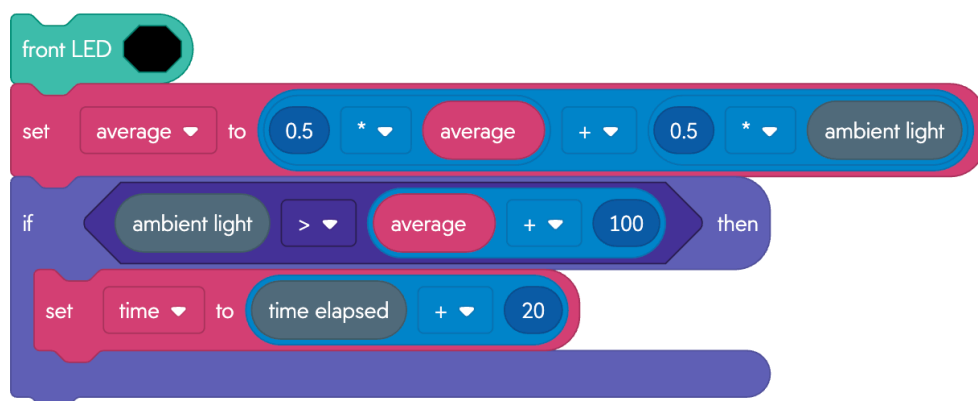
13) In **Variables**, create a new variable called **average**.

This is a **number** as it records the average light level.

14) Add the code (below) at the end of the loop (still inside the loop) to take the average:

$(0.5 * \text{average}) + (0.5 * \text{ambient light})$.

If the brightness increases by 100 then set **time** to 20 seconds in the future, during which Pac-Man is **energized**.



To **Save** your code go back to the main page. Don't forget to add "Pac-Man" to the name of your program. Click on the 3 dots (ellipsis) next to your program and download to your computer.